SONGA TRACK

INTERNAL · PRODUCT & ENGINEERING STRATEGY

Tracking & Security, as a Business

Songa's live-tracking engine is already production-grade. This proposal spins it out as a standalone B2B service — transit-cargo visibility, vehicle anti-theft, and fleet telematics for customers who never book a single Songa trip.

Document	Prepared	Model	Core risk
2 of 2 — Songa Track	**20 July 2026**	**B2B SaaS · per-asset**	**Low — engine exists**

Same codebase, separate business. Reuses the Redis telemetry → Kafka → Java WebFlux gateway → partitioned-Postgres breadcrumb → deck.gl H3 live-map stack that already powers Songa trips today.

Songa built a real-time location platform to run its trucking marketplace: sub-second driver telemetry in Redis, a Kafka bus, a Java Spring WebFlux WebSocket gateway that manages H3-based location groups, month-partitioned GPS breadcrumb history in PostGIS, polyline/ETA/traffic reconstruction, and a deck.gl H3 heat-map admin. **That engine is the product.** Songa Track sells it to third parties — logistics firms, cargo owners, and vehicle owners — as tracking-and-security-as-a-service, **decoupled from the trip.**

TRACKING CORE	CORE RISK	MODEL	NEW SURFACE
Shipped	**Low**	**Per-asset**	**Effort L**
Telemetry, breadcrumbs, ETA, H3 live-map already run in production	The hard real-time part exists; new work is packaging & access	Recurring per-vehicle / per-shipment / month SaaS revenue	Multi-tenant API, device abstraction, geofence/alerts, portal

Why this is a credible standalone business

The asset is real and rare. A working, low-latency, H3-indexed fleet-tracking pipeline with tamper-evident history is expensive to build. Songa already paid that cost for its marketplace. Reselling it is near-pure margin on sunk engineering.

The market is adjacent. The same Tanzanian/EA logistics and vehicle-owner customers Songa already touches need cargo-in-transit visibility and anti-theft — problems the trip engine already solves internally.

It de-risks the marketplace. Tracking revenue is recurring and counter-cyclical to trip GMV — it earns whether or not a shipment is booked, smoothing the P&L.

It has a natural wedge. Start with Songa's own fleet-owners (Doc 1, line #1) as design partners, then sell outward to non-Songa fleets.

The one hard boundary: today, location is meaningful only *inside a Trip*. Songa Track's central engineering task is to make an **Asset** (a truck, a shipment, a private car, a hardware GPS tracker) a first-class tracked entity *without* a Songa trip. Everything else is packaging around the existing pipeline.

What Already Exists — The Engine

The tracking pipeline that runs Songa trips today, end to end. Every box below is shipped code.



ENGINE CAPABILITY	STATUS	WHERE IT LIVES
Live position + heading + H3 index per moving asset (Redis)	EXISTS	`Driver.last_known_location/h3_index`, Redis state
Real-time push to viewers over WebSocket, H3-grouped	EXISTS	`songa_ws_gateway/` (Spring WebFlux), `utils/kafka.py`
Month-partitioned GPS breadcrumb history at scale	EXISTS	`TripLocationHistory` (psqlextra partitioning)
Route reconstruction: polylines, ETA, traffic-delay & idle-stop analytics	EXISTS	`TripActivitySummary`, `services/google_maps_service.py`
H3 supply-density / demand heat-maps, live-asset map	EXISTS	`admin_api/views/live`, `songa_admin …/live-map` (deck.gl + maplibre + h3-js)
Tamper-evident event history (hash-chained ledger pattern)	EXISTS	`payments/models/LedgerEntry.py` (reusable pattern for audit trail)
SMS + WS notification delivery	EXISTS	`services/nextSMS.py`, Kafka broadcast
Asset tracked **without** a Trip (standalone device/shipment)	MISSING	Location is Trip-scoped today
Multi-tenant API + API-key / customer isolation	MISSING	No external tenant model
Geofence + alert-rule engine (enter/exit/idle/speed/route-deviation)	MISSING	No rules engine
Webhooks / customer-facing tracking dashboard	MISSING	Admin map is internal-only
3rd-party hardware GPS-tracker ingestion (non-phone devices)	MISSING	Ingest is phone/driver-app only

Read this table as the whole thesis: the top seven rows — the genuinely hard, latency-critical, at-scale parts — already ship. The bottom five are conventional API/product work. That asymmetry is why core risk is **Low** even though the surface area is **Effort L**.

Three use-cases, one engine. Each is the same telemetry pipeline pointed at a different asset type and wrapped in different alerts.

① Transit-Cargo Visibility

Shippers & cargo owners watch a consignment move door-to-door with ETA, route history and delay alerts.

- Live shipment location + ETA
- Route-deviation & long-idle alerts
- Tamper-evident delivery history
- Shareable tracking link for their customer

ENGINE READY

② Vehicle Security / Anti-Theft

Vehicle owners get geofences, movement alerts and last-known-location recovery for cars & trucks.

- Geofence enter/exit alerts
- Unexpected-movement / tow alert
- Speed & night-use alerts
- Location history for recovery / insurance

+ GEOFENCE ENGINE

③ Fleet Telematics (non-Songa)

External fleets that never book Songa trips use the dashboard + API for utilisation & driver behaviour.

- Fleet-wide live map (deck.gl)
- Utilisation, idle-time, mileage reports
- API + webhooks into their systems
- Per-vehicle history & scoring

+ API + DEVICES

Reuse vs. Build (whole business)

✓ REUSE — THE EXPENSIVE 80%

- Redis live-state + H3 indexing pipeline
- Kafka → Java WebFlux WS gateway (JWT, H3 groups) — `songa_ws_gateway/`
- Partitioned breadcrumb history — `TripLocationHistory`
- Polyline / ETA / traffic-delay / idle analytics — `TripActivitySummary`
- deck.gl + maplibre + h3-js live map — `songa_admin …/live-map`
- Hash-chained audit pattern for tamper-evidence — `LedgerEntry`
- SMS/WS notification delivery — `services/` , Kafka

▲ BUILD — THE CONVENTIONAL 20%

- `Asset` + `Device` models (track without a Trip)
- `Tenant` + API-key / OAuth, per-tenant data isolation
- Public REST + webhook API & SDK
- Geofence + `AlertRule` engine (enter/exit/idle/speed/route-deviation)
- Customer-facing tracking dashboard + shareable links
- Hardware GPS-tracker ingestion adapter (e.g. TCP/MQTT device protocols)
- Billing per active asset / month (reuse invoice/ledger)

Key data-model changes

CHANGE	TYPE	NOTES
`Tenant(name, api_key, plan, isolation)`	New	External customer; scopes all data
`Asset(tenant, type, label, current_h3, last_seen)`	New	Truck / shipment / car / device — the thing tracked, no Trip required
`Device(asset, kind, identifier, protocol)`	New	Phone app OR hardware tracker feeding the asset
Generalise breadcrumb history from `Trip` → `Asset`	Refactor	Reuse partitioning; add `asset_id` alongside `trip_id`
`Geofence(tenant, polygon/h3set)` + `AlertRule(asset, type, params)`	New	enter/exit/idle/speed/route-deviation
`AlertEvent` (hash-chained)	New (reuse pattern)	Tamper-evident, insurance-grade audit
`Webhook(tenant, url, events)`	New	Push alerts into customer systems

The pivotal refactor is decoupling location history from `Trip`. Because breadcrumbs are already partitioned and written from Redis streams, adding an `asset_id` dimension is an additive migration, not a rewrite — the ingestion path stays the same.

Packaging & pricing (illustrative)

Track
per asset / month · entry

- Live location + ETA
- 7-day history
- Shareable tracking link
- Basic idle/stop alerts
- Email/SMS notifications

Secure
per asset / month · popular

- Everything in Track
- Geofences + enter/exit/speed alerts
- 90-day tamper-evident history
- Anti-theft / movement alerts
- Webhooks

Fleet / API
volume + platform fee

- Everything in Secure
- Full REST API + SDK
- Fleet live map + reports
- Hardware-tracker ingestion
- SLA + priority support

Prices intentionally omitted — set from device/data cost + competitive per-asset benchmarks during the pilot. Model is recurring per active asset, with volume tiers for fleets and metered API/overage.

Revenue model

RECURRING
Per-asset
Predictable MRR independent of trip GMV

EXPANSION
Tier + assets
Land on Track, expand to Secure/Fleet & more vehicles

HARDWARE
Device margin
Optional GPS-tracker resale + activation

Go-to-market wedge

PHASE 1 · DESIGN PARTNERS

Own fleet

Songa's fleet-owners (Doc 1 #1) track their trucks — free/beta, proves value

PHASE 2 · ADJACENT

Shippers

Cargo owners already on Songa buy transit visibility for all shipments

PHASE 3 · OUTWARD

Non-Songa

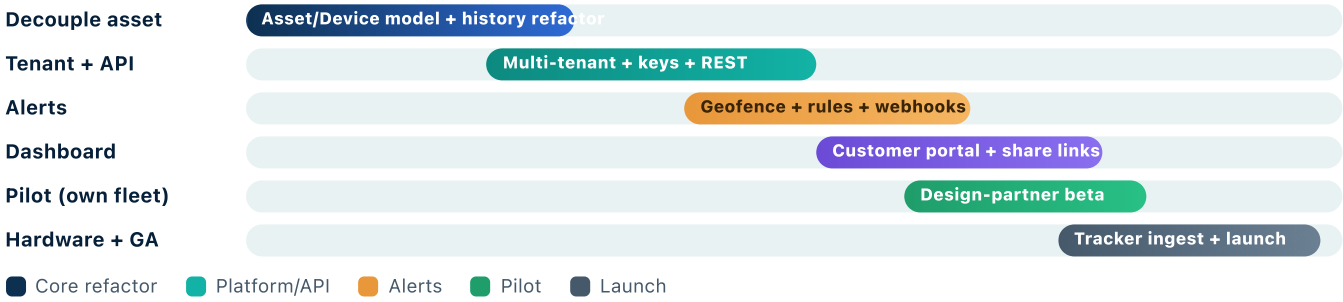
Sell to fleets & vehicle owners who never book a trip

Security, privacy & compliance

- **Tenant isolation** — API keys + row-level scoping; a tenant only ever sees its own assets.
- **Tamper-evident history** — reuse the hash-chained ledger pattern so location/alert logs are insurance- and dispute-grade.
- **Consent & data protection** — vehicle/person tracking is sensitive; enforce owner consent, retention limits, and align to Tanzania's Personal Data Protection Act. Legal review required before public launch.
- **Auth** — the WS gateway already validates JWT and manages H3 groups; extend to tenant-scoped tokens.

Primary risks: (1) **Regulatory** — tracking people/vehicles carries privacy obligations; treat consent + retention as launch-blocking, not fast-follow. (2) **Hardware** — supporting third-party GPS trackers means device-protocol integration & support load; keep Phase 1 phone-app-only. (3) **Support expectation** — security customers expect 24/7 reliability & SLAs, a step up from marketplace ops.

Roadmap



Bottom line

Songa Track monetises the platform's single strongest, already-built asset with low core-engineering risk. The genuinely hard real-time tracking pipeline is done and in production; the remaining work — an asset/tenant abstraction, an alert engine, and a customer portal — is conventional product build. Sequenced behind Doc 1's fleet-owner line as design partners, it becomes a recurring-revenue business that also de-risks marketplace GMV. The gating concern is not engineering but **privacy/regulatory compliance**, which must be resolved before any public launch.